

A071 · Interpretable and scalable spatial gene set activity analysis with GESSO uncovers functional tissue architecture

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Recent advances in spatially resolved transcriptomics (SRT) enabled measurement of sets of pathway genes activity within tissues. However, existing gene set activity scoring methods overlook spatial dependencies among tissue locations, restricting their ability to capture region-specific pathway activities associated with disease pathology or cellular communication. Moreover, these methods lack inference on statistical significance for activity scores, provide limited interpretability of gene-level contribution to a pathway, and scale poorly to advanced large-scale SRT datasets. To address these limitations, we present GESSO (Gene sEt activity Score analysis with Spatial lOcation), a spatially informed gene set scoring method adaptable to diverse SRT platforms. GESSO models gene set activity levels through a graph-regularized matrix decomposition algorithm, jointly inferring spatially coherent gene set activity scores (GASs) and interpretable gene contributions that quantify how strongly individual genes drive the GASs. It further implements a permutation-based local significance test and a stratified low-resolution approximation that scales to high-resolution SRT datasets such as Visium HD, Stereo-seq, and Xenium Prime. Across 13 datasets from five SRT platforms, GESSO outperformed all existing methods in accuracy, calibration, interpretability, and scalability. Applications revealed novel biological programs, including spatially confined EMT activation within tumor-stroma interfaces, developmental signaling gradients across embryonic tissues, and coordinated B-cell, T-cell, and signaling pathways within germinal centers of human lymph node tissue, revealing the spatial organization of immune function at subregional resolution.